

Aortic Aneurysm & Dissection Quiz

Faculty text edition • Version 1.0

Format

10 questions | Five choices (A-E) | Answer, explanation, pearl, image reference

Question 1

Which layer of the aortic wall first tears in an acute aortic dissection?

- A. Adventitia
- B. Intima
- C. Media
- D. Endothelium only
- E. Vasa vasorum

Correct answer: B

Explanation: An intimal tear allows blood to enter the media, creating a false lumen.

Educational pearl: Hypertension is the most common risk factor for aortic dissection.

Gallery image reference: 3

Question 2

Which classification system is most commonly used for aortic dissections?

- A. Wells
- B. Stanford
- C. Rutherford
- D. CEAP
- E. Fontaine

Correct answer: B

Explanation: Stanford classifies dissections as Type A (ascending aorta involved) or Type B (ascending aorta not involved).

Educational pearl: Type A dissections usually require emergency surgery.

Gallery image reference: 1

Question 3

Which symptom is most characteristic of acute aortic dissection?

- A. Gradual calf pain
- B. Sudden severe tearing chest or back pain
- C. Chronic cough
- D. Fever
- E. Abdominal bloating

Correct answer: B

Explanation: Patients classically describe abrupt onset of severe tearing or ripping pain.

Educational pearl: Pain migration may indicate propagation of the dissection.

Gallery image reference: 4

Question 4

Which imaging study is preferred for a hemodynamically stable patient with suspected aortic dissection?

- A. Chest X-ray
- B. Echocardiography only
- C. CT angiography
- D. Plain CT
- E. MRI brain

Correct answer: C

Explanation: CTA rapidly defines the location and extent of dissection and is the preferred initial study in stable patients.

Educational pearl: CTA is also excellent for evaluating aneurysm size.

Gallery image reference: 6

Question 5

What is the major goal of initial medical management for acute aortic dissection?

- A. Increase heart rate
- B. Lower blood pressure and reduce shear stress
- C. Administer antibiotics
- D. Begin thrombolysis
- E. Correct anemia

Correct answer: B

Explanation: Anti-impulse therapy with beta-blockers and blood pressure control reduces propagation of the dissection.

Educational pearl: Heart rate control is often achieved before aggressive blood pressure reduction.

Gallery image reference: 7

Question 6

Which inherited disorder is strongly associated with thoracic aortic aneurysm?

- A. Cystic fibrosis
- B. Marfan syndrome
- C. Crohn disease
- D. Psoriasis
- E. Multiple sclerosis

Correct answer: B

Explanation: Marfan syndrome predisposes patients to medial degeneration and thoracic aortic aneurysm.

Educational pearl: Young patients with thoracic aneurysms should be evaluated for genetic syndromes.

Gallery image reference: 8

Question 7

Which physical examination finding may occur in proximal (Type A) aortic dissection?

- A. Carpal tunnel syndrome
- B. Diastolic murmur of aortic regurgitation
- C. Splenomegaly
- D. Clubbing
- E. Wheezing

Correct answer: B

Explanation: Extension into the aortic root can produce acute aortic regurgitation.

Educational pearl: Pulse deficits increase suspicion for dissection.

Gallery image reference: 5

Question 8

Which complication is most feared with an enlarging abdominal aortic aneurysm?

- A. Venous thrombosis
- B. Rupture
- C. Appendicitis
- D. Pancreatitis
- E. Diverticulitis

Correct answer: B

Explanation: Rupture carries a very high mortality and is the principal indication for elective repair based on aneurysm size.

Educational pearl: Larger aneurysms have a greater annual rupture risk.

Gallery image reference: 7

Question 9

Which treatment is definitive for most Stanford Type A dissections?

- A. Aspirin
- B. Observation
- C. Emergency surgical repair
- D. Physical therapy
- E. Long-term antibiotics

Correct answer: C

Explanation: Type A dissections are surgical emergencies because of the risk of rupture, tamponade, and coronary involvement.

Educational pearl: Mortality rises rapidly if surgical treatment is delayed.

Gallery image reference: 9

Question 10

What is the most important long-term management strategy after successful treatment of aortic aneurysm or dissection?

- A. Lifelong blood pressure control and surveillance imaging
- B. Daily aspirin alone
- C. Bed rest
- D. High-intensity exercise
- E. Routine antibiotics

Correct answer: A

Explanation: Strict blood pressure control and periodic imaging are essential to monitor for disease progression or recurrence.

Educational pearl: Lifelong follow-up is required even after successful surgical or endovascular repair.

Gallery image reference: 10